value distribution and thus can learn the distribution of observed rewards rather than just the expected value. This is useful for the reasons we described in the first section—by learning a distribution we have a way to utilize risk-sensitive policies that take into consideration the variance and possible multimodality of the distribution.

7.4. Distributional Q-learning

We've now covered all the preliminaries necessary to implement a distributional deep Q-network (Dist-DQN). If you didn't completely understand all of the material in the previous sections, don't worry; it will become more clear when we start writing the code.

In this chapter we are going to use one of the simplest Atari games in the OpenAI Gym, Freeway (figure 7.7), so that we can train the algorithm on a laptop CPU. Unlike other chapters, we're also going to use the RAM version of the game. If you look at the available game environments at https://gym.openai.com/envs/#atari, you will see that each game has two versions, with one being labeled with "RAM."

Figure 7.7. Screenshot from the Atari game Freeway. The objective is to move the chicken across the freeway, avoiding oncoming traffic.